

BHABHA UNIVERSITY, BHOPAL

NH-12, Hoshangabad Road, Jatkhed, Bhopal-462026

POST GRADUATE DIPLOMA IN COMPUTER APPLICATIONS (PGDCA)



DETAILED SYLLABUS (EFFECTIVE FROM JULY 2018)

**Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL**

BHABHA UNIVERSITY, BHOPAL
POST GRADUATE DIPLOMA IN COMPUTER APPLICATIONS
(PGDCA)

(EFFECTIVE FROM JULY 2018)
DETAILED SYLLABUS

SCHEME

SEMESTER –II

S.No.	Paper Code	Paper Name	Internal Evaluation	Theory	Practical	Total
1	PGDCA21	Software Engineering	30	70		100
2	PGDCA22	Web Technologies	30	70		100
3	PGDCA23	RDBMS(Oracle)	30	70		100
4	PGDCA24	Data Communication and networks	30	70		100
PRACTICAL						
5	PGDCA25	Lab Oracle	20		30	50
6	PGDCA26	Lab Web Technologies	20		30	50
TOTAL MARKS			160	280	60	500

PAPER CODE – PGDCA21

Software Engineering

UNIT-1

Software: Characteristics, Components, Applications, And Software Process Models: Waterfall, Spiral, Prototyping, Fourth Generation Techniques, Concepts of Project Management, Role of Metrics &Measurements.

UNIT-2

S/W Project Planning: Objectives, Decomposition techniques: S/W Sizing, Problem-based estimation, Process based estimation, Cost Estimation Models: COCOMO Model.

UNIT-3

The S/W Equation, System Analysis: Principles of Structured Analysis, Requirement analysis, DFD, Entity Relationship diagram, Data dictionary.

UNIT-4

S/W Design: Objectives, Principles, Concepts, Design methodologies: Data design, Architectural Design, procedural design, Object -oriented concepts

UNIT-5

Testing fundamentals: Objectives, principles, testability, Test cases: White box & Black box testing, Testing strategies: verification & validation, unit test, integration testing, validation testing, system testing.

Text & Reference Books:

1. Roger. S. Pressman, Software Engineering - A Practitioner's Approach, 7th Edition, McGraw Hill, 2010.
2. Rajib Mall, “Fundamental of Software Engineering “, 3rd edition, PHI, 2009.
3. Naseeb Singh Gill, “Software Engineering: Software reliability, testing and quality, Khanna Book Publishing, 2011.

PAPER CODE – PGDCA22

Web Technologies

UNIT-1

Introduction to HTML : Information Files Creation; Web Server; Web Client/Browser (Understanding how a Browser communicates with a Web Server); Hyper Text Markup Language (HTML Tags, Paired Tags); Commonly used HTML Commands Titles and Footers; Text Formatting (Paragraph Breaks, Line Breaks); Emphasizing Material in a Web Page (Heading Styles, Drawing Lines); Text Styles (Bold, Italics, Underline); Other Text Effects (Centering (Text, Images etc.); Spacing (Indenting Text). Adding Graphics to HTML Documents: Using the Border attribute; using the Width and Height Attribute; Using the Align Attribute; Using the ALT Attribute.

UNIT-2

Tables : Introduction (Header, Data rows, The Caption Tag); Using the Width and Border Attribute; Using the Cell padding Attribute; Using the Cell spacing Attribute; Using the BGCOLOR Attribute; Using the COLSPAN and ROWSPAN Attributes. Linking Documents: Links External Document References, Internal Document References

UNIT-3

Introduction to JavaScript : JavaScript in Web Pages (Netscape and JavaScript, Database Connectivity, Client side JavaScript, Capturing User Input); The Advantages of JavaScript (An Interpreted Language, Embedded within HTML, Minimal Syntax - Easy to Learn, Quick Development, Designed for Simple, Small Programs, Performance, Procedural Capabilities, Writing JavaScript into HTML.

UNIT-4

Forms Used by a Web Site: The Form Object; The Form Object's Methods (The Text Element, The Password Element, The Button Element, The Submit (Button) Element, The Reset (Button) Element, The Checkbox Element, The Radio Element, The Text Area Element, The Select and Option Element, The Multi Choice Select Lists Element)

UNIT-5

Objects in JavaScript (The String Object, The Math Object, The Date Object); User Defined Objects (Creating a User Defined Object, Instances, Objects within Objects). Cookies: What are Cookies; Setting a Cookie.

Text & Reference Books:

1. Internet for EveryOne: Alexis Leon, 1st Edition, Leon Techworld, Publication, 2009.
2. Greenlaw R; Hepp E, "Fundamentals of Internet and WWW" 2nd Edition, Tata McGraw-Hill
3. Raj Kamal, "Internet & Web Technologies" edition Tata McGraw-Hill Education.2009

PAPER CODE – PGDCA23

RDBMS(Oracle)

UNIT-1

An Overview of DBMS and DB Systems Architecture : Introduction to Database Management systems, Data Models, Database System Architecture, Relational Database Management systems, Candidate Key and Primary Key in a Relation, Foreign Keys, Relational Operators, Set Operations on Relations, Attribute domains and their Implementation.

UNIT-2

The Normalization Process : Introduction, first Normal Form, Partial Dependencies, Second Normal Form, data Anomalies in 2NF Relations, Transitive Dependencies, Third Normal Form.

UNIT-3

The Entity Relationship Model: The Entity Relationship Model, Entities and Attributes, Relationships, One-One Relationships, Many-to-one Relationships, Normalizing the Model, Table instance charts. Interactive SQL : SQL commands , Data Definition Language Commands, Data Manipulation Language Commands, insertion of data into the tables, Viewing of data into the tables, Deletion operations, updating the contents of the table, modifying the structure of the table, renaming table, destroying tables.

UNIT-4

Viewing The Data : Computations on Table Data, Arithmetic Operators, Logical Operators, Comparison Operators, Range Searching, Pattern Searching, ORACLE FUNCTIONS, Group Functions, Scalar Functions, Data Conversion Functions, Manipulating Dates in SQL .

UNIT-5

Sub queries and Joins : Joins, Equi Joins, Non Equi Joins, Self Joins, Outer Joins, Sub Queries, Correlated Queries, Using Set Operators:- Union , Intersect, Minus. Views and Indexes : Definition and Advantages Views, Creating and Altering Views, Using Views, Indexed Views, Partitioned views, Definition and Advantages of Indexes, Composite Index and Unique Indexes, Accessing Data With and without Indexes, Creating Indexes and Statistics.

Text & Reference Books:

- 1.Ramez Elmasri, "Fundamentals of Database Systems", Pearson Education India, 2009
- 2.JD Ullman,Garcia-Molina, "Database System: The Complete Book", Edition 4th, Pearson
- 3.S.K Singh, "Database Systems: Concepts, Design and Applications", Edition 2nd, Pearson
- 4 C.J Date, "An Introduction to Database System", Edition 8th, Pearson Education India. 2009

PAPER CODE – PGDCA24

Data Communication and networks

UNIT-1

Introduction to data communication, analog Vs Digital Communication, Fourier Analysis, Band Width limitation, data rate of a channel, Error detection and correction; nature of errors, parity check, CRC, hamming code, Modulation; Multiplexing: SDM, FDM, TDM, STDM.

UNIT-2

Introduction to computer networks and application; network hardware, network software, OSI reference model, TCP/IP model, network standardization, physical layer: circuit switching, packet switching, message switching, terminal handling, telephone system, modems, connections.

UNIT-3

Transmission media, Data link layer: design issues, elementary data link protocols-sliding window protocol, HDLC/SDLC, ALOHA, CSMA/CD, token passing, IEEE standard 802 for LAN and WAN. Network layer: design issues, Routing algorithms: shortest path routing.

UNIT-4

Congestion control algorithms: leaky bucket, token bucket, Internet working, the network layer in the Internet IP protocol, IP address.

UNIT-5

Transport layer: design issues, elements of transport protocol, addressing establishing & releasing a connection, flow control & buffering, TCP/IP service model, TCP connection management.

Text & Reference Books:

- 1 Andrew S. Tanenbaum, “Computer Networks”, 3rd Edition, Pearson Prentice Ltd. 2010.
- 2 Behrouz A Forouzan, “Data Communication and Networking” 4th Edition, Tata McGraw Hill,
- 3 Larry L. Peterson, “Computer Networks: A System Approach”, 4th Edition, Elsevier Publication